

## MAT3032 ANSWERS TO EXERCISES 2.5

### Exercises 2.5

1. (a)  $75 = 3 \times 5^2$  so  $G$  has  
 3-Sylows of order 3:  $w_3 \equiv 1 \pmod{3}$  and  $w_3 \mid 25$ , so  $w_3 = 1$  or  $25$ .  
 5-Sylows of order 25:  $w_5 \equiv 1 \pmod{5}$  and  $w_5 \mid 3$ , so  $w_5 = 1$ .  
 (b)  $126 = 2 \times 7 \times 3^2$  so  $G$  has  
 2-Sylows of order 2:  $w_2 \equiv 1 \pmod{2}$  and  $w_2 \mid 63$ , so  $w_2 = 1, 3, 7, 9, 21$  or  $63$ .  
 3-Sylows of order 9:  $w_3 \equiv 1 \pmod{3}$  and  $w_3 \mid 14$ , so  $w_3 = 1$  or  $7$ .  
 7-Sylows of order 7:  $w_7 \equiv 1 \pmod{7}$  and  $w_7 \mid 18$ , so  $w_7 = 1$ .  
 (c)  $175 = 5^2 \times 7$  so  $G$  has  
 5-Sylows of order 25:  $w_5 \equiv 1 \pmod{5}$  and  $w_5 \mid 7$ , so  $w_5 = 1$ .  
 7-Sylows of order 7:  $w_7 \equiv 1 \pmod{7}$  and  $w_7 \mid 25$ , so  $w_7 = 1$ .  
 (d)  $400 = 2^4 \times 5^2$  so  $G$  has  
 2-Sylows of order 16:  $w_2 \equiv 1 \pmod{2}$  and  $w_2 \mid 25$ , so  $w_2 = 1, 5$  or  $25$ .  
 5-Sylows of order 25:  $w_5 \equiv 1 \pmod{5}$  and  $w_5 \mid 16$ , so  $w_5 = 1$  or  $16$ .
2.  $133 = 19 \times 7$ . As 7 does not divide  $19 - 1$ , by Proposition 2.23 there is no non-abelian group of order 133. Thus such a group must be abelian, so by Proposition 2.22 (ii) it has one 7-Sylow. (Or find  $w_7$  as in Q.1.)
3.  $|G| = 45 = 3^2 \times 5$  so  $G$  has  
 3-Sylows of order 9:  $w_3 \equiv 1 \pmod{3}$  and  $w_3 \mid 5$ , so  $w_3 = 1$ .  
 5-Sylows of order 5:  $w_5 \equiv 1 \pmod{5}$  and  $w_5 \mid 9$ , so  $w_5 = 1$ .  
 Hence by Proposition 2.22 (i) the 3-Sylow  $H$  and the 5-Sylow  $J$  are normal in  $G$ . By Proposition 2.22 (iii),  $G \cong H \times J$ .  
 By Proposition 2.8,  $H$  is isomorphic to  $C_3 \times C_3$  or  $C_9$ . As  $|J|$  is prime,  $J \cong C_5$ .  
 Thus the only possibilities for  $G$  are  $C_3 \times C_3 \times C_5$  and  $C_9 \times C_5$ . Clearly these are abelian.
4.  $\sigma^2 = (1\ 3\ 5\ 7\ 2\ 4\ 6)$  so (as in Example 2.3.3)  $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 3 & 5 & 7 & 2 & 4 & 6 \end{pmatrix} = (2\ 3\ 5)(4\ 7\ 6)$ .  
 We have  $\sigma^7 = \tau^3 = \iota$  so  $|\langle \sigma \rangle| = 7$ ,  $|\langle \tau \rangle| = 3$ . As 3 and 7 are coprime,  $|\langle \sigma \rangle \langle \tau \rangle| = 21$  so  $\langle \sigma \rangle \langle \tau \rangle = G$ .  
 $\tau\sigma = \sigma^2\tau \neq \sigma\tau$  so  $\langle \sigma, \tau \rangle$  is a non-abelian group of order 21. It has presentation  $\langle \sigma, \tau : \sigma^7 = \tau^3 = \iota, \tau\sigma\tau^{-1} = \sigma^2 \rangle$ .  
 (We found a non-abelian group of order 21 in Examples 2.5.3 (ii). As this is unique, the two groups must be isomorphic.)

5.  $711 = 3^2 \times 79$ . The group of units of  $\mathbb{Z}_{79}$  has order 78 and is cyclic, so it has a cyclic subgroup of order 3 (since  $3 \mid 78$ ). Let this subgroup be  $H = \{1, g, g^2\}$ ; for example, we could take  $g = 23$  and then  $g^2 = 55, g^3 = 1$  (but you don't need to find  $g$  to know it exists).

Consider the set of all matrices of the form  $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$  where  $a, c \in \{1, g, g^2\}$  and  $b \in \mathbb{Z}_{79}$ . This has order  $3 \times 3 \times 79 = 711$  and is easily seen to be a non-abelian subgroup of  $GL_2(\mathbb{Z}_{79})$ .

There are two abelian groups of order 711, namely  $C_3 \times C_3 \times C_{79}$  and  $C_9 \times C_{79}$ , so there are at least three non-isomorphic 711-element groups.

In the above group of matrices,  $w_{79} = 1$  so there is a normal subgroup of order 79. It consists of the matrices of the form  $\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$ .

6.  $w_7 \equiv 1 \pmod{7}$  and  $w_7 \mid 33$ , so  $w_7 = 1$ . Thus  $G$  has a unique 7-Sylow  $J$ , of order 49, which is normal in  $G$ .

$|G/J| = 33 = 3 \times 11$ . As  $3 \nmid (11 - 1)$ ,  $G/J$  is cyclic.

By Proposition 1.8,  $G$  has a subgroup  $H$  isomorphic to  $G/J$ , i.e. to  $C_{33}$ .

Suppose this is normal in  $G$ .  $\varphi(33) = 20$ , which is coprime to 1617, so by Proposition 2.24  $H \subset Z(G)$ .

As  $|H|$  and  $|J|$  are coprime,  $H \cap J = \{e\}$  and  $|HJ| = 33 \times 49$  so  $HJ = G$ .

As  $H \triangleleft G$  and  $J \triangleleft G$ , by Proposition 1.5 (ii) we have  $G \cong H \times J$ .

7. A subgroup of order 7 is a 7-Sylow. As it is unique, it is normal in  $G$  so the quotient group  $G/N$  exists and has order  $3^2 \times 5 = 45$ .

Applying the Sylow theorems to  $G/N$  we find  $w_3 = 1$  so  $G/N$  has a unique, hence normal, 3-Sylow of order 9. By the subgroup correspondence theorem this subgroup must be  $H/N$  where  $H$  is a normal subgroup of  $G$ , of order  $9 \times 7 = 63$ .

8. Let  $|G| = 455 = 5 \times 7 \times 13$ .  $w_{13} \equiv 1 \pmod{13}$  and  $w_{13} \mid 35$ . Thus  $w_{13} = 1$  so  $G$  has exactly one subgroup  $H$  of order 13, which is normal in  $G$ .

$w_7 \equiv 1 \pmod{7}$  and  $w_7 \mid 65$ . Thus  $w_7 = 1$  so  $G$  has exactly one subgroup  $J$  of order 7, which is normal in  $G$ .

Thus  $HJ$  is a group and  $H, J$  are both normal in  $HJ$ . Also  $H \cap J = \{e\}$  as  $|H|$  and  $|J|$  are coprime. Thus  $HJ \cong H \times J = C_{13} \times C_7 \cong C_{91}$ .  $[G : HJ] = 5$ , the smallest prime factor of  $|G|$ , so  $HJ \triangleleft G$ .

Let  $K$  be a 5-Sylow.  $|K| = 5$  and  $\varphi(91) = 72$  are coprime, so all elements of  $HJ$  commute with all elements of  $K$  by Proposition 2.24. Also  $HJ \cap K = \{e\}$  as  $|HJ|$  and  $|K|$  are coprime, so  $HJK \cong HJ \times K \cong C_{91} \times C_5 \cong C_{455}$ .

Then clearly  $HJK = G$ , so  $G \cong C_{455}$ . Thus all groups of order 455 are cyclic.

9.  $w_p \equiv 1 \pmod{p}$  and  $w_p \mid (p^2 - 1) = (p - 1)(p + 1)$ .

Hence  $w_p = 1$  or  $p + 1$ . In the latter case, by Proposition 2.21 (v)  $G$  has a subgroup of index  $p + 1$ , which has order  $|G|/(p + 1) = p^2 - p$ .

10. (a)  $w_3 \equiv 1 \pmod{3}$  and  $w_3 \mid 2$  so  $w_3 = 1$ . Thus  $G$  has a unique, hence normal, 3-Sylow of order  $3^2 = 9$ .

$G$  also has at least one 2-Sylow, which must have the form  $\{e, a^2\}$  where  $a^2 = e$ , i.e.  $o(a) = 2$ .

- (b) If  $N = \langle b \rangle = \{e, b, \dots, b^8\}$  then, as  $N$  is closed under conjugation,  $aba^{-1} \in N$  so  $aba^{-1} = b^r$  where  $r \in \{0, \dots, 8\}$ .

$b = ebe = a^2ba^{-2} = a(aba^{-1})a^{-1} = ab^ra^{-1} = (aba^{-1})^r$  by Proposition 1.2 (ii), so  $b = (b^r)^r = b^{(r^2)}$ .

Thus  $r^2 \equiv 1 \pmod{9}$  so  $r = 1$  or  $r = 8$ . If  $r = 1$  then  $ab = ba$ . But  $e, b, \dots, b^8, a, ab, \dots, ab^8$  are the 18 distinct elements of  $G$ , so if  $ab = ba$  then  $G$  is abelian. Hence  $r = 8$  and  $aba^{-1} = b^8 = b^{-1}$ , so  $bab = a$ .

$G = \langle a, b : a^2 = b^9 = e, bab = a \rangle$  (or equivalent).  $G \cong D_9$ .

- (c) An automorphism of  $G$  is determined by its effect on the generators  $a$  and  $b$ , each of which must map to an element of the same order.  $G$  has nine elements of order 2 (the elements  $ab^k$ ) and six of order 9 (all powers of  $b$  except  $b^0, b^3, b^6$ ), so the maximum possible order of  $\text{Aut}(G)$  is  $9 \times 6 = 54$ .

Now suppose  $\phi : G \rightarrow G$  is defined on the generators by  $\phi(a) = ab^k$ ,  $\phi(b) = b^\ell$  where  $k \in \{0, \dots, 8\}$ ,  $\ell \in \{1, 2, 4, 5, 7, 8\}$ , and extended to make it a homomorphism, i.e.  $\phi(ab^m) = \phi(a)\phi(b)^m = (ab^k)(b^\ell)^m = ab^{k+\ell m}$ .

Then  $\phi(ab^m) \neq e$  since  $a \notin \langle b \rangle$ , and  $\phi(b^m) = e$  iff  $b^{\ell m} = e$ , i.e.  $9 \mid \ell m$ , which is not the case for any of the possible values of  $m$ . Hence  $\phi$  has a trivial kernel, so it is an automorphism for any  $k, \ell$ . Thus  $|\text{Aut}(G)| = 54$ .

From Exercises 1.5 Q.4,  $G/Z(G) \cong \text{Inn}(G)$ . As  $|Z(G)| = 1$  (given) we have  $|\text{Inn}(G)| = |G| = 18$ .